

DRONE DEVELOPMENT CHALLENGE 2024

DESIGN REPORT



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DDC2024079

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APPENDIX A - STATEMENT OF COMPLIANCE

SAEISS DRONE DEVELOPMENT CHALLENGE 2024

Certification of Qualification

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Statement of Compliance

As Faculty Advisor:

- I certify that the registered team members are enrolled in collegiate courses.
- I certify that this team has designed, constructed and/or modified the radio-controlled airplane with the intention to use this aircraft in the **SAEISS Drone Development Challenge 2024** competition, without direct assistance from professional engineers, R/C model experts or pilots, or related professionals.
- I certify that this year's Design Report has original content written by members of this year's team.
- I certify that all reused contents have been properly referenced and is in compliance with the University's plagiarism and reuse policies.


Signature of Faculty Advisor

19/02/2024
Date


Signature of Team Captain

19/02/2024
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Note: A signed scanned copy of the statement should be included in your Design Report as Page 2

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List of Abbreviation

SAEISS = Society of Automotive Engineers India, Southern Section	b = Wingspan
DDC = Drone Design Challenge	S_{HT} = Horizontal Tail Area
UAV = Unmanned Aerial Vehicle	S_{VT} = Vertical Tail Area
FRP = Fibre Reinforced Plastic	X_{HT} = Length between the aerodynamic centres of the wing and horizontal stabiliser
MAC = Mean Aerodynamic Chord	X_{VT} = Length between the aerodynamic centres of the wing and vertical stabiliser
Vn Diagram = Speed vs Load Factor Chart	$\bar{c}$ = Wing mean aerodynamic chord
CL = Coefficient of Lift	Li-Po = Lithium Polymer
CD = Coefficient of Drag	ESC = Electronic Speed Controller
CM = Coefficient of Moment	BEC = Battery Eliminator Circuit
AOA = Angle of Attack	T/W = Thrust to weight Ratio
SW = Wing Area	n = Load Factor
W/S = Wing Loading	MDF = Medium Density Fiber
EPS = Expanded Polystyrene	

1. ABSTRACT

Aircraft design is a complex process that involves the creation of new aircraft from the initial concept to the final product. This process includes 4 major stages: Conceptual Design, Preliminary Design, Detailed Design, and Testing. Aerodynamics, Structures, Systems, and Propulsion are just a few of the disciplines that go into designing an aircraft. To design an aircraft that meets the criteria of given mission requirements, additional considerations like safety, performance, efficiency, etc. are also taken into consideration. The technical data, design process, results of performance analysis, and test results of the project are all

documented in the report that follows. Also, adequate attention has been made in predicting the payload, which can aid us in gaining valuable competition points. As recent technology and improvements are introduced, aircraft design techniques continue to change, and it is essential to the growth of the aviation and transportation sectors.

2. EXECUTIVE SUMMARY

The Society of Automotive Engineers conducts SAEISS Drone Development Challenge 2024. Our team, VOYAGERS, is formed to represent MIT ADT University, Pune in the SAEISS DDC 2024. The SAEISS Drone Development Challenge aims to promote and develop Indian expertise and experience in unmanned systems technologies at the university and college levels. The competition provides two categories for the teams to choose from for competing. These categories are Regular Class and Micro Class.



Figure 01: 3-D render of designed UAV

Formed under the Laggar Falcon Aero Club, Team Voyagers has registered to compete in the Regular Class of the competition. Team Voyagers was formed to learn and explore the concepts of aeromodelling. The objective of Regular Class is to design a fixed-wing UAV maximum 72 inches wingspan. Designing such an aircraft is a challenge, as maximum lifting capacity must be achieved while keeping in mind the design restrictions and the overall weight of the aircraft.

The design constraints are as follows:

- Fully configured for take-off, the free-standing aircraft shall have a maximum combined length, width, and height (L+W+H) of 150 inches. The wingspan should not exceed 72 inches.
- Aircraft may not weigh more than 4kgs and not less than 2kgs.

- The use of Reinforced Plastic (FRP) is prohibited on all parts of the aircraft.

The following report contains all the technical data, analysis results, design decisions, test results and performance analysis of the aircraft designed by our team for the competition.

3. WING DESIGN

3.1 AIRFOIL SELECTION

The initial, critical stage in aircraft design is choosing the airfoil. Airfoils that produce high lift, but less drag are favoured. Also, the aircraft's stability is greatly influenced by the coefficient of moment of the airfoil. Hence, our group considered three airfoils with different cambers and thicknesses. The following is a list of the three airfoils:

1.) SG6043:

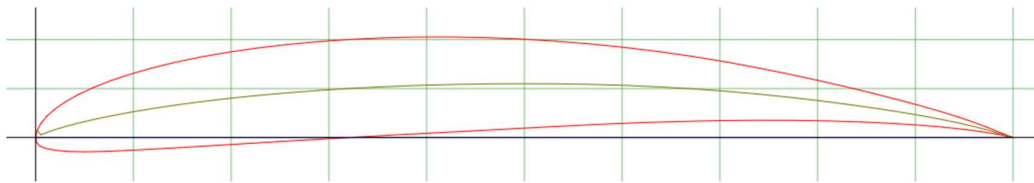


Figure 02: Cross Section of SG 6043 Airfoil

2.) NACA 4412:

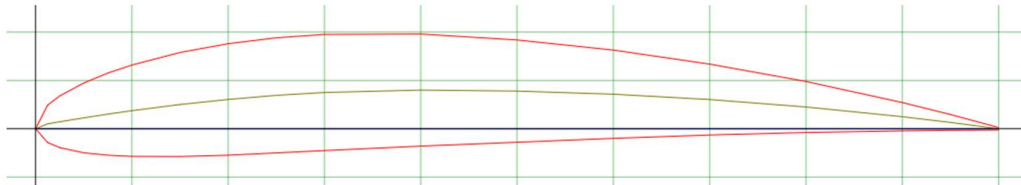


Figure 03: Cross Section of NACA 4412 Airfoil

3.) ISA 962:

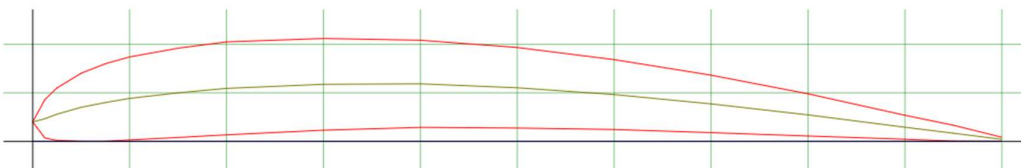


Figure 04: Cross Section of ISA 962 Airfoil

The comparison between the specifications of the various airfoils is stated in the table below:

Sr. No.	Airfoil	Specifications
1	SG6043	Maximum thickness 10% at 32.1% chord Maximum camber 5.1% at 53.3% chord
2	NACA 4412	Maximum thickness 12% at 30% chord Maximum camber 4% at 40% chord
3	ISA 962	Maximum thickness 9.6% at 20% chord Maximum camber 4.6% at 40% chord

Table 01: List of airfoils analysed.

Analysis was conducted for varying Reynolds number from an operating range of 30000 to 500000.

The above results are for Reynolds number 160000 as these airfoils are distinguished for its low-speed characteristics suitable for our aircraft application which operates in a range of 10-15m/s.

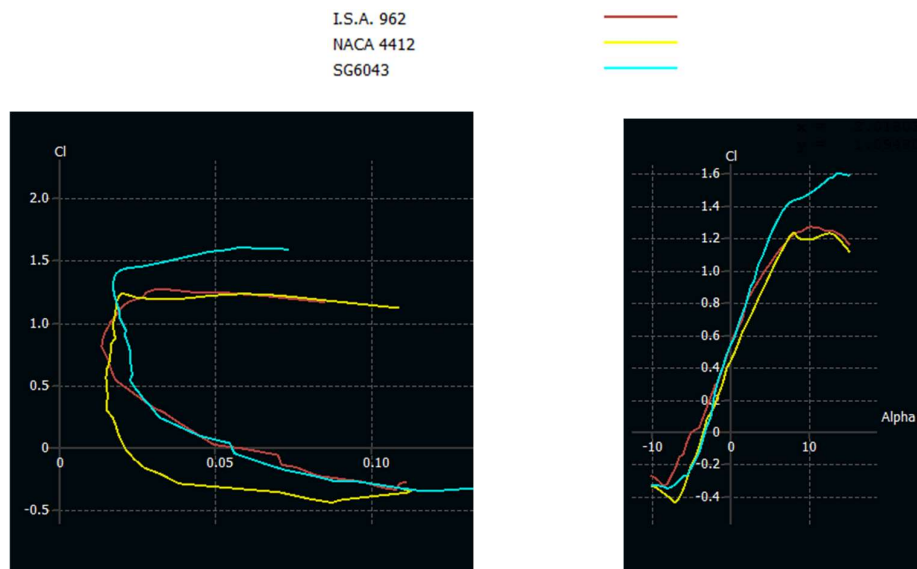


Figure 05: Coefficient of Lift vs Coefficient of Drag **Figure 06:** Coefficient of Lift vs Alpha

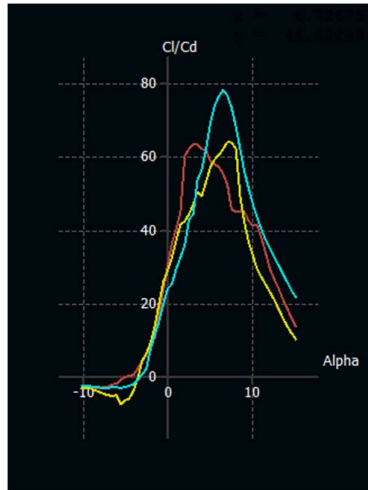


Figure 07: C_l/C_d vs Alpha

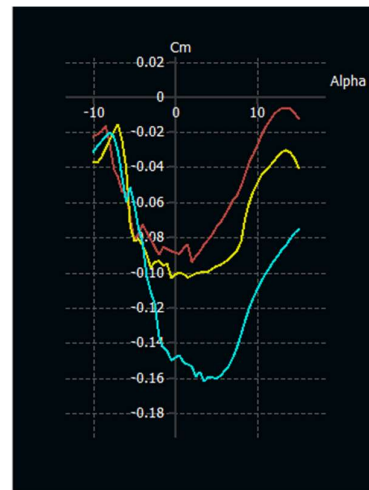


Figure 08: Coefficient of moment vs Alpha

- SG 6043 has the highest Coefficient of lift-to-Coefficient of drag ratio compared to other airfoils, with a maximum of about 1.6.
- Both ISA 962 and SG 6043 have a gentle stall slope, with SG 6043 having the largest stall angle of around 13.
- The C_d produced by ISA962 is the lowest of the three, followed by NACA4412.
- SG6043 has the least coefficient of moment and ISA 962 has the highest coefficient of moment.

Despite having a greater c_d than NACA 4412 and ISA 962, we chose SG 6043 because of its high C_l/C_d and C_l values, which also provide a higher stall angle.

3.2 WING CONFIGURATION

3.2.1 Wing Position

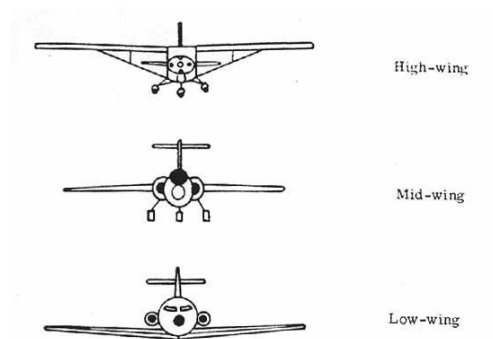


Figure 09: Wing Positions

High Wing	There is more space to carry more payload and it is easier to load or unload the aircraft.
Low Wing	The aircraft take-off performance is better due to the ground effect.
Mid Wing	It is aerodynamically more streamlined when compared with the other two configurations.

Table 02: Comparison Between Different Wing Positions

Our team decided on a high wing layout for our aircraft since it allows for the transportation of a larger cargo, which is the requirement for the competition.

3.2.2 Wing Shape

Deciding the wing shape is one of the most crucial parts of making an aircraft. There are 3 types of wing shapes, each one with their own disadvantages and advantages. Those are as follows:

Rectangular Wing	These wings are easier and cheaper to construct.
Tapered Wing	In addition to being more structurally efficient, tapered wings are more aerodynamically effective than constant-chord wings.
Elliptical Wing	They have the same effective AOA over the whole span because they provide the same amount of lift throughout the entire wing.

Table 03: Wing Shapes

We have chosen a rectangular wing, since it is cost effective and easier to manufacture. Using the fundamental wing lift equation and accounting for the aircraft's expected lift weight, the planform area was computed.

3.3 WING SIZING

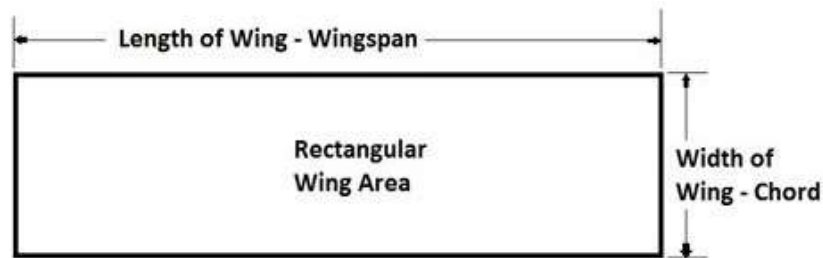


Figure 10: Wing Sizing Specifications

o Wingspan- 68.46 inches

o Wing chord- 13.4 inches

3.3.1 Surface Area Calculation

o Span= 63.74 inches (excluding fuselage),

o Chord= 13.6 inches

o Planform Area for Rectangular Wing = SPAN*CHORD = $13.6 * 63.74 = 866.864 \text{ inch}^2$

3.3.2 Wing Loading

Wing loading is the ratio of an aircraft's weight to its wing area, and it is denoted as W/S. This parameter shows the amount of load, or weight, that each wing unit can hold. When adopting high wing loading, lesser payload carrying capacities can be obtained at higher flight speeds, even supersonic flight speeds. On the other hand, flights with slower speeds but more payload carrying capacity and manoeuvrability are recommended to have less wing loading. Aircraft with low wing loads need longer runways for take-off and landing. Aircraft with low wing loading can operate at slower take-off and landing speeds without encountering any issues. Consequently, we try to calculate and minimise wing loading to build an aircraft that can carry maximum payload at lower flying speeds and be easily manoeuvrable.

Wing loading calculation: -

Total take-off weight of aircraft (estimated) = 19.84 lbs

Planform area = 866.864 inch^2

Wing loading = Take-off weight/Wing Area = 16.330 kg/m^2

3.3.3 Control Surfaces of the wing

Control surfaces are aerodynamic devices that help manoeuvre the plane or adjust the attitude of the plane during flight. The control surfaces we used on the plane are as follows:

- **Ailerons:** Ailerons, a basic flight control surface, restrict motion around the longitudinal axis of an aircraft. This motion is referred to as a roll.

In the design process of an aileron, four parameters need to be determined. They are:

- (i) aileron planform area (S_a),
- (ii) aileron chord/span (C_a/b_a),
- (iii) maximum up and down aileron deflection ($\pm\delta_{Amax}$),
- (iv) location of inner edge of the aileron along the wingspan (b_{ai}).

As a general guidance, the typical values for these parameters are as follows: $S_a/S = 0.05-0.1$, $b_a/b = 0.2-0.3$, $C_a/C = 0.15-0.25$, $b_{ai}/b = 0.6-0.8$, and $\delta_{Amax} = \pm 30$ deg. Based on this, about 5–10% of the wing area is devoted to the aileron, the aileron-to-wing-chord ratio is about 15–25%, the aileron-to-wing-span ratio is about 20–30%, and the inboard aileron span is about 60–80% of the wing span. Hence taking above characteristics:

- o Aileron area(A)= 41.52 inch²
- o Chord length of aileron(l)= 2.68 inches
- o Span of aileron(b)= 15.55 inches

- **Flaps:** A flap is a high-lift device used to reduce an aircraft wing's stalling speed at a specific weight. In a fixed-wing aircraft, flaps are typically positioned on the trailing edges of the wings. Flaps are used to shorten the distances required for take-off and landing.

$$S_{\text{flap}} = \frac{2 \cdot \Delta L}{\rho \cdot V^2 \cdot e_f \cdot C_{L_{\text{flap}}}}$$

Figure 11: Surface Area of a Flap

The incremental lift (ΔL) due to the flaps can be expressed as the difference between the lift with flaps (L_{flap}) and the lift without flaps (L_{wing}):

$$\Delta L = L_{\text{flap}} - L_{\text{wing}} = 0.9$$

The flap efficiency factor (e_f) represents the effectiveness of the flaps in generating lift relative to the additional drag produced:

$$e_f = \Delta D / \Delta L = 0.85$$

Hence,

$$\text{Flap area (A)} = 41.52 \text{ inch}^2$$

$$\text{Chord length of Flap (l)} = 2.68 \text{ inches}$$

$$\text{Span of Flap (b)} = 15.55 \text{ inches}$$

3.4 WING CONSTRUCTION

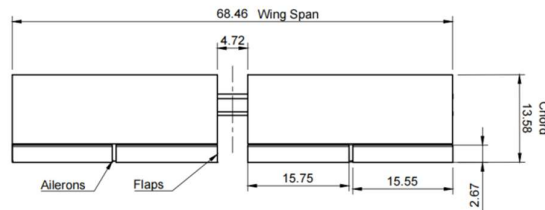


Figure 12: Wing Planform



Figure 13: Wing Construction

4. FUSELAGE

The fuselage houses the crucial components that bring your aircraft to life. It is the backbone holding the wings, tail, and landing gear, payload and provides space for the motor, battery, receiver, and servos.

4.1 SHAPE

To reduce induced drag and increase lift, our team chose to build the fuselage using an airfoil. To create the fuselage, our team adjusted the AH 93 airfoil.

Fuselage length is 21.25 inches.

4.2 TAIL POSITIONING

The position of an aircraft's tail is significant because the arm length, or the distance from 25% of the wing to 25% of the tail, is considered when calculating the horizontal and vertical tail's surface area. The moment arm of our plane is *31.65 inches*.

4.3 DRAG ANALYSIS

The primary goal of the analysis is to determine the aircraft design's zero lift drag, or the aerodynamic drag coefficient of the design when the wings are not producing any lift. The analysis will also pinpoint any potential trouble spots, such as designs like the fuselage-wing junction, which negatively impact the aircraft's aerodynamic performance.

4.4 FUSELAGE CONSTRUCTION

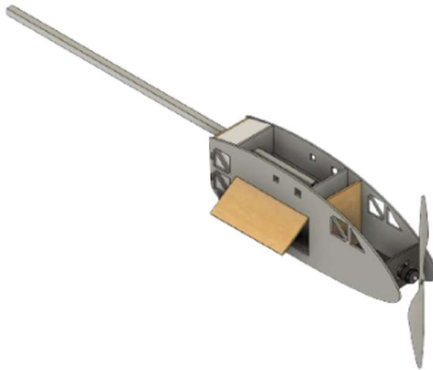


Figure 14: Fuselage Construction

5. EMPENNAGE

It is the structure at the back of an aeroplane that provides stability during flight is the empennage, commonly referred to as the tail. In addition to providing control surfaces for the aircraft's pitch and yaw motion, tail surfaces enhance overall aircraft stability. While developing these surfaces, extra care is taken because they increase the aircraft's structural weight and wetted area.

5.1 EMPENNAGE CONFIGURATION

A U-tail design, with two vertical stabilizers placed at the ends of horizontal stabilizer, it provides better effective control and stability than conventional tail as the wider placement of the vertical stabilizers can offer improved yaw stability.

$$S_{HT} = \frac{c_{HT} \bar{c}_W S_W}{X_{HT}} \quad S_{VT} = \frac{c_{VT} b_W S_W}{X_{VT}}$$

Figure 15: Volumetric method Formula

Horizontal tail

-Area, $S_{HT} = 151.5 \text{ inch}^2$

-Volumetric Coefficient, $C_{HT} = 0.5$

-Moment arm, $X_{HT} = 31.65 \text{ inches}$

-Aspect Ratio = 2.5

-Span, $b_{HT} = 19.46 \text{ inches}$

-MAC = 7.784 inches

-Root chord, $C_r = 10.38 \text{ inches}$

-Taper Ratio = 0.5

-Tip Chord, $C_t = 5.184 \text{ inches}$

Vertical tail

-Area, $S_{VT} = 15.15 \text{ inch}^2$

-Volumetric Coefficient, $C_{VT} = 0.04$

-Moment arm, $X_{VT} = 31.65 \text{ inches}$

-Aspect Ratio = 2

-Span, $b_{VT} = 5.5 \text{ inches}$

-MAC = 2.9 inches

-Root chord, $C_r = 3.75 \text{ inches}$

-Taper Ratio = 0.6

-Tip Chord, $C_t = 2.24 \text{ inches}$

5.2 CONSTRUCTION

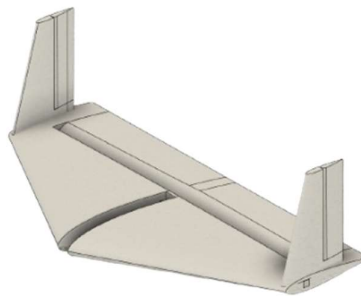


Figure 16: Tail Construction

6. LANDING GEAR

The landing gear supports the aircraft during landing and while it is on the ground. There are two basic configurations of aeroplane landing gear: conventional gear or tail wheel gear and the tricycle gear. The tail dragger landing gear system, also known as conventional landing

gear, is a classic aircraft design featuring two main wheels positioned forward of the centre of gravity and a single, smaller wheel (or skid) at the tail. In Tricycle landing gear, two main wheels are located behind the centre of gravity and a single, smaller nose wheel is located in the front.

Our Team chose tail dragger landing gear configuration as they can be safely used in rough terrains, are very efficient in short field take-off, comparatively lighter and are easy to construct. Also, this landing gear configuration is best suited for our empennage design considering the competition design constraints.



Figure 17: Main landing Gear



Figure 18: Tail landing Gear

7. PROPULSION AND ELECTRICAL SYSTEM

A battery, propeller, and electric motor are all components of an electric propulsion system. Electric propulsion is used in most of the RC Aircraft (wingspan less than 2 m). Electric propulsion systems' key distinguishing characteristic is that they are best suited for aircraft with a mass of less than 30 kg. A typical electric-powered aircraft also has the advantages of being inexpensive, small, having a constant centre of gravity, and being silent.

7.1 THRUST CALCULATION

The preliminary step required to choose an electrical propulsion system is the calculation of thrust. Typically for a RC Aircraft, the thrust-to-weight ratio is between 0.35 and 0.55. Considering, a thrust-to-weight ratio of 0.45, the required thrust can be calculated as:

Thrust Required = Weight * Thrust-to-Weight Ratio

$$\text{Thrust} = 9,000\text{g} * 0.45 = \mathbf{4050\text{g}}$$

7.2 MOTOR SELECTION

Choosing an appropriate motor and propeller can determine the entire flight of the aeroplane. Depending upon the thrust value as calculated before, a kV value is selected. A lower kV value is selected, as preference is given to carrying more payload than the speed of the aircraft.

7.2.1 Power and Thrust Calculations

- Empty weight $\sim 4 \text{ kg} = 8.82 \text{ lbs}$ - Total weight $\sim 9 \text{ kg} = 19.84 \text{ lbs}$
The Inputs Watts Per Pound lies between 110 - 130 watts per pound. Hence, a value of 130 was chosen to calculate power required.
- Empty: $P_e = 8.82 \times 130 = 1146.6 \text{ W}$
- Total: $P_t = 19.84 \times 130 = 2579.2 \text{ W}$
- Required: $P = \text{avg. } (P_e \text{ \& } P_t) = 2009.8 \text{ W} \sim 2000 \text{ W}$

Hence, Power required = 2000 W

Generally, Thrust/Weight ratio lies between 0.6 - 0.8.

Assumed T/W = 0.7

Thrust, $T = 0.7 \times \text{Total weight} = 0.7 \times 9 = 6.3 \text{ kg}$

Accordingly, the following motors were shortlisted by the team:

Model Name	T-Motor AT 5220-A	Avian 6362	SunnySky X5212S
Image			
Max Power	2200 W	1900 W	800W
Motor kV	380 kV	200 kV	280 kV
Max Current	100 A	100 A	46A

Max Input Voltage	6 S	12 S	6S
Cost	\$ 139.00	\$ 179.99	\$ 79.99
Weight	0.465 kg	0.635 kg	0.28 kg
Thrust @ 100% Throttle	7.806 kg	9.040 kg	4.82 kg
Propeller used	(18*10)''	(19*10)''	(22*6.6)''

Table 04: Motor Comparison

7.3 BATTERY SELECTION

With the motor known, a suitable battery was researched. Per the rules of the competition, it was known that the battery had to be a commercially available 6S Lithium-Polymer battery. With the power known (power limited to 2000 W by the calculations) and knowing that the battery must be 6S (22.2 V), it was determined that the maximum required current is 90.00 A. Also, it must be able to deliver a minimum of 51.64 A. With these constraints, Orange 2500 mAh 6S 35C/70C (22.2V) Lithium Polymer Battery Pack (7*2*1.5)'' was chosen.

7.4 ESC SELECTION

An electronic speed controller or ESC is a device installed to a remote-controlled electrical model to vary its motor's speed and direction. It needs to plug into the receiver's throttle control channel. We have used 115A electronic speed controllers to control each brushless motors in this experiment which can constantly supply required current to drive brushless motors. It has following specifications:

Input Voltage	3S-6S LiPo
BEC Voltage	6 V or 8 V, user selectable; 9 A continuous, 20 A peak output

Table 05: ESC Specifications

7.5 SERVO SIZING

The selection of a suitable servo motor for an RC aircraft is a crucial aspect of the design process that requires careful consideration of several factors. The primary consideration is the torque required to move the control surface of the aircraft, which is dependent on the size and weight of the aircraft, as well as the type of control surface being used. To determine the required torque, the control surface area, control surface deflection, and air pressure must be taken into account.

$$\text{Torque} = \text{Control surface area} \times \text{Control surface deflection} \times \text{Air pressure}$$

Where,

- control surface area is the surface area of the control surface in square inches
- control surface deflection is the angle of deflection in degrees
- air pressure is the air pressure in pounds per square inch (psi)

For this particular servo motor sizing, considering the flap dimensions and the maximum angle of deflection as 30°, the required torque was calculated as,

$$\text{Torque} = 41.52 \times 30 \times 14.7$$

$$\text{Torque} = 1.8 \text{ kg/cm}$$

This torque value obtained was used to find a servo motor with the necessary specifications to produce the required torque. Thus, it was found out that the SG90 servo provides an optimum torque in the range of 1.8 kg-cm to 2.2 kg-cm. Similar calculations were carried out on the other control surfaces and with their required torque lying within the provided range.

Furthermore, the size and weight of the servo motor was also considered. This servo is only 13.4g in weight, and thus will not negatively impact the balance and stability of the aircraft and will fit in the available space.

8. ANALYSIS

8.1 STABILITY ANALYSIS AND CONTROL

The control system of the drone movement has six degrees of freedom, consisting of three translational and three rotational movements, which are crucial for manoeuvring stability. Longitudinal stability and control are concerned with the aircraft's pitching motion. Lateral stability and control are concerned with the aircraft's rolling motion and directional stability and control are related to the aircraft's yawing motion.

The aircraft is said to be stable if it returns to its original equilibrium state after a small perturbing displacement without any external involvement. Thus, the aircraft's response arises purely from the inherent design. At level flight, we tend to refer to this as static stability.

In effect the aeroplane is statically stable when it returns to the original steady flight condition after a small disturbance; statically unstable when it continues to move away from the original steady flight condition upon a disturbance; and neutrally stable when it remains steady in a new condition upon a disturbance.

The aeroplane may converge continuously back to the original steady flight state; it may overcorrect and then converge to the original configuration in an oscillatory manner, or it can diverge completely and behave uncontrollably. This kind of stability is associated with dynamic stability.

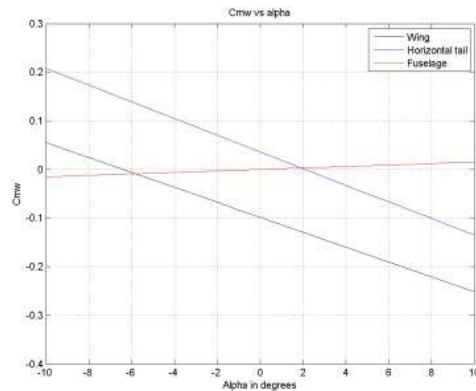


Figure 19: Longitudinal Stability

Elevator: Control surface on the horizontal stabilizer of the empennage, plays a crucial role in controlling the aircraft's pitch.

In the design of the elevator, four parameters should be determined. They are:

- (i) elevator planform area (S_E)
- (ii) elevator chord (C_E)
- (iii) elevator span (b_E)
- (iv) maximum elevator deflection ($\pm\delta_{E_{max}}$)

As a general guidance, the typical values for these parameters are as follows:

$S_E/S_h = 0.15-0.4$, $b_E/b_h = 0.8-1$, $C_E/C_h = 0.2-0.4$, $\delta_{E_{max_up}} = -25^\circ$ and $\delta_{E_{max_down}} = +20^\circ$.

Hence by above references the calculation is:

o Chord length of elevator = *1.37 inches*

o Area of elevator = *24.2216 inch²*

o Span of elevator = *17.68 inches*

Rudder: Control surface on the vertical stabilizer of the empennage, governs the yaw of the aircraft.

In the design of the rudder, four parameters must be determined:

- (i) rudder area (S_R),
- (ii) rudder chord (C_R),
- (iii) rudder span (b_R),
- (iv) maximum rudder deflection ($\pm\delta_{Rmax}$),

Hence,

- o Chord length of rudder = *3.89 inches*
- o Area of rudder = *18.478 inch²*
- o Span of rudder = *4.75 inches*

8.2 STATIC THRUST ANALYSIS

We investigated the performance of a specific motor with varying propeller sizes and pitches. The aim was to understand how changes in propeller configurations influence thrust and RPM. Through a series of static thrust tests, we explored the relationship between thrust and RPM for different propeller setups. Our analysis revealed insights into optimal operating points, showcasing the trade-offs between thrust and power consumption.

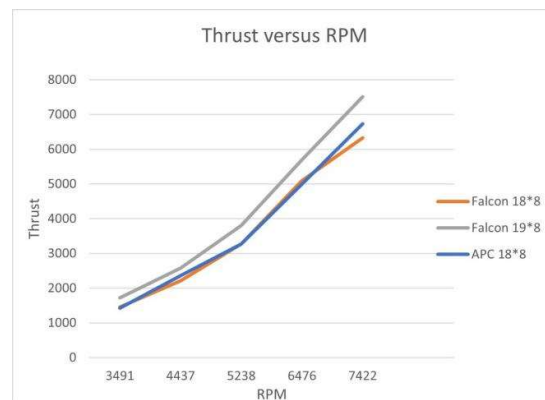


Figure 20: Thrust vs RPM

8.3 STRUCTURAL ANALYSIS

Boundary Conditions:

- Force applied
 - For Landing Gear=110 N
 - For Wing to Fuselage= 55N
- Fixed Support applied at required positions.

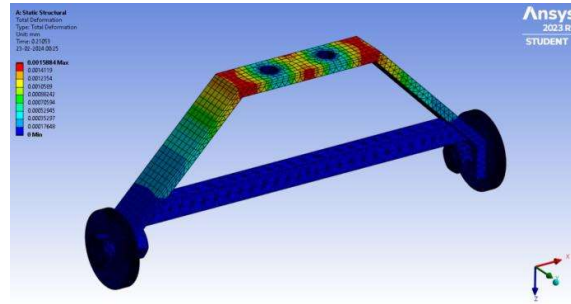


Figure 21: Landing Gear Deformation

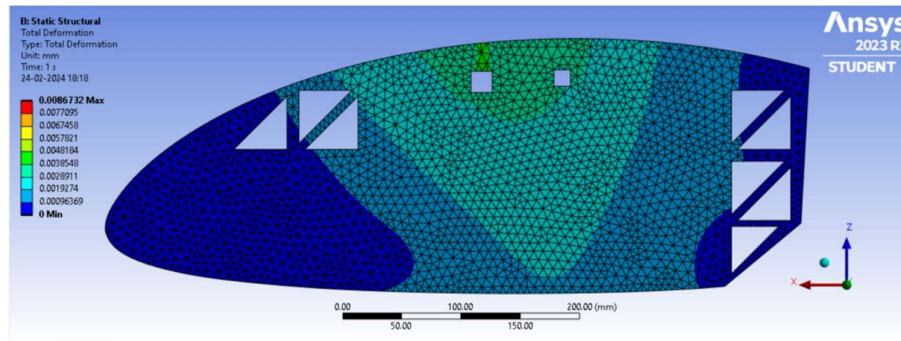


Figure 22: Wing to Fuselage Deformation

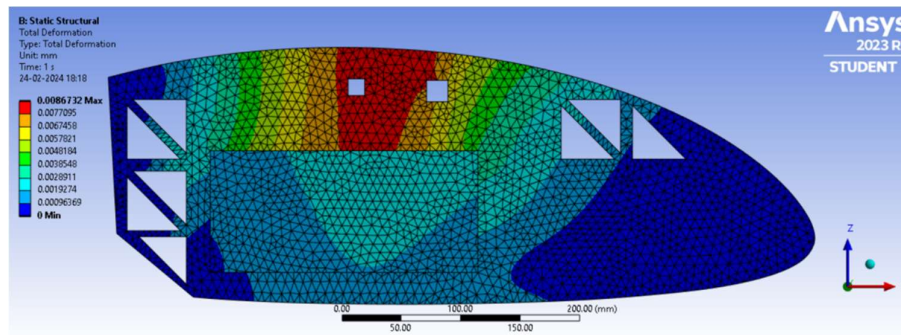


Figure 23: Wing to Fuselage (Hatch Opening) Deformation

8.4 PERFORMANCE ANALYSIS

8.4.1 *V-n* Diagram

The *V-n* diagram is made to estimate the structural limitations of our aircraft. The lift/weight ratio or load factor, *n* varies from aircraft to aircraft depending on its application. It was found that for our aircraft, the recommended load factor lies between 2 and 3 and if this factor range is exceeded, the aircraft will risk its structural integrity. The following graph shows the

relationship, so that it may be studied for any speed and the maximum load factor can be established for it.

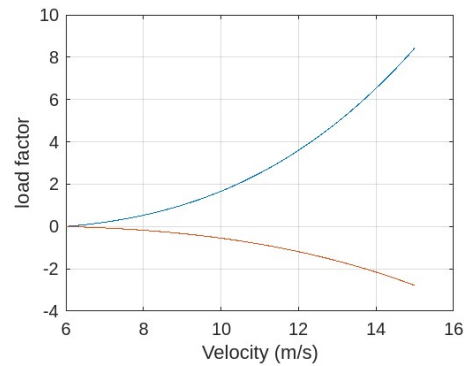


Figure 24: Vn Diagram

8.5 AERODYNAMIC ANALYSIS

- Boundary Conditions:

Inlet velocity: 15m/s

Viscous Model: k-omega SST

8.5.1 2-D Analysis over selected Airfoil

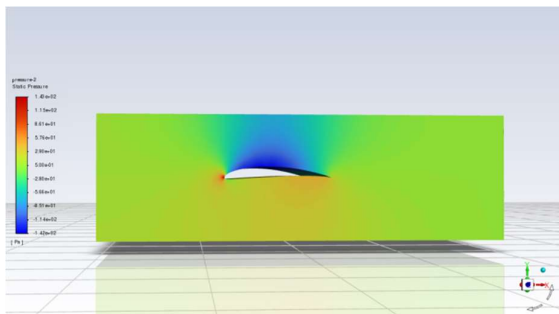


Figure 25: Pressure contour

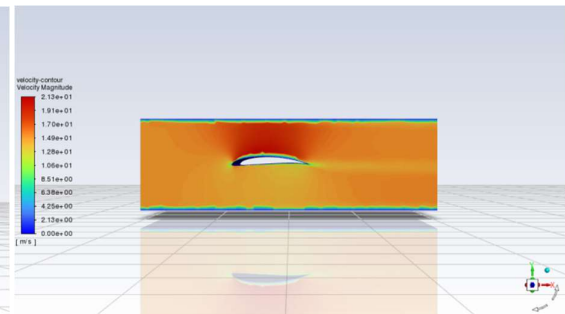


Figure 26: Velocity contour

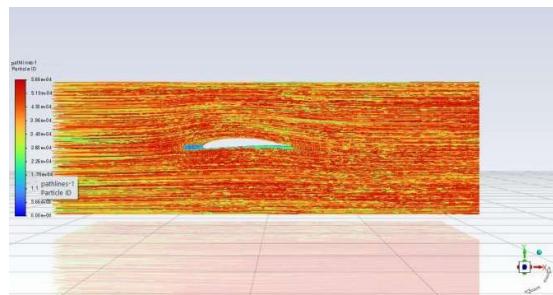


Figure 27: Streamlines

8.5.2 3-D Analysis over Aircraft

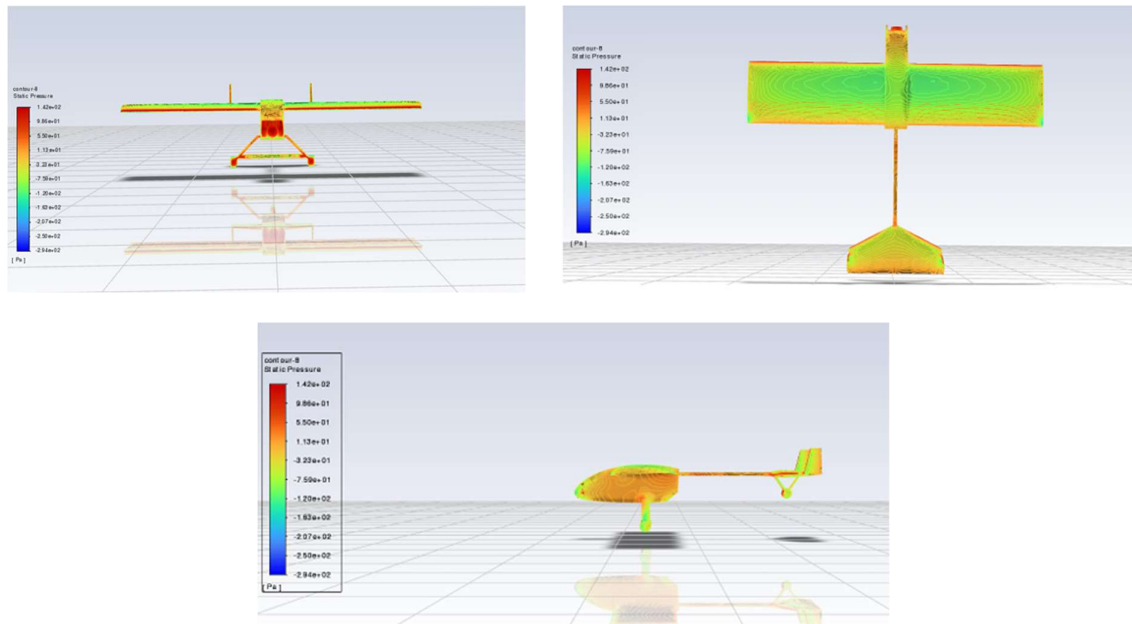


Figure 28: Pressure Distribution Over Aircraft Front view, Top View, Side view

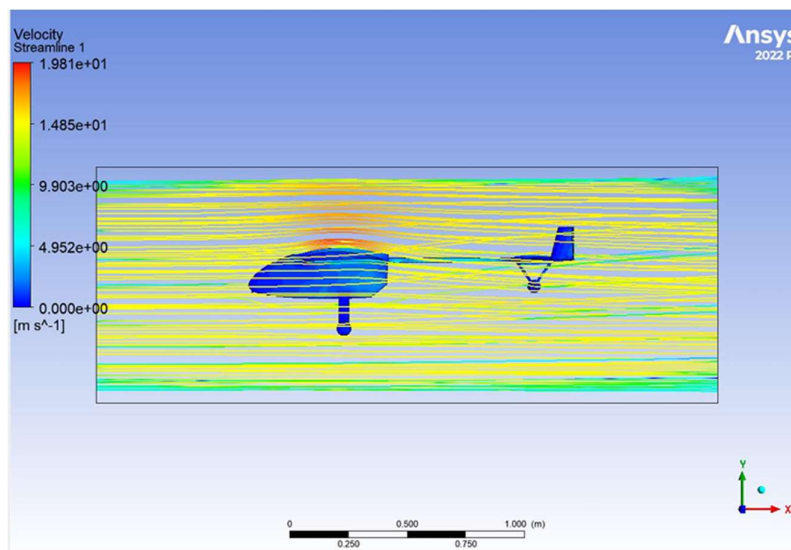


Figure 29: Velocity Streamlines Over Aircraft Side view

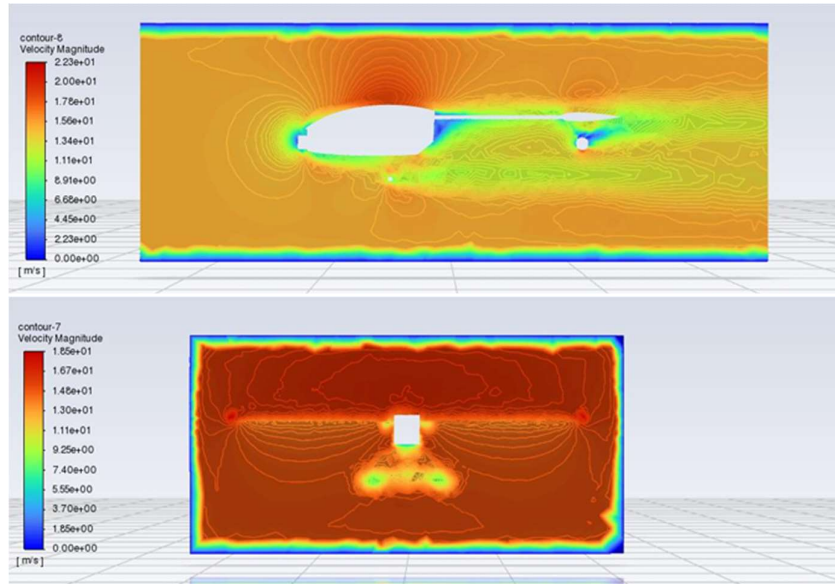


Figure 30: Velocity Contours Over Aircraft Side view, Front view

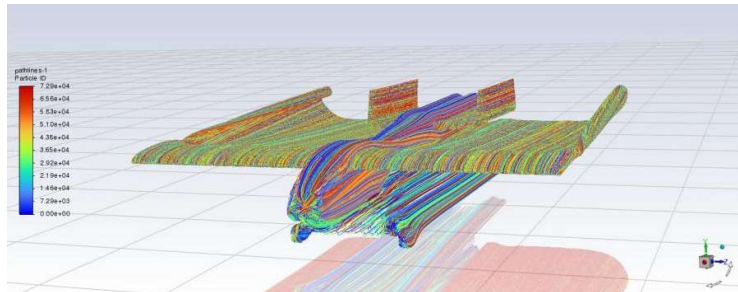


Figure 31: Velocity pathlines over the aircraft

9. PAYLOAD PREDICTION

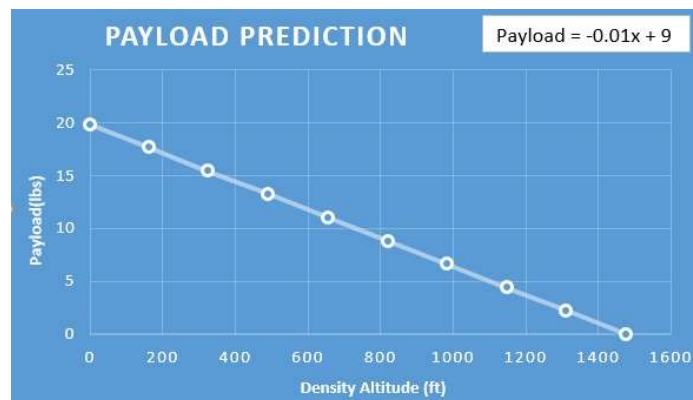


Figure 32 : Payload Prediction Curve

$$\text{Endurance} = \text{Battery capacity} / \text{Power consumption rate}$$

10. MANUFACTURING PROCESS

The manufacturing process of an RC plane involves a systematic approach, starting with the design and material selection. The wing of our plane is made of Expanded Polystyrene (EPS), chosen for its lightweight properties. The wing cores are precision-cut using a hot wire cutter. The fuselage, responsible for housing the plane's components, is constructed from lightweight and durable materials such as balsa wood and Medium Density Fiber (MDF). The tail section, including the horizontal and vertical stabilizers, is also made from EPS. The landing gear, crucial for take-off and landing, is made of aluminium for strength and reduced weight. The careful selection of materials for each component ensures a balance between structural integrity and weight, resulting in an RC plane with optimal performance characteristics.

11. CONCLUSION

The design of the RC aircraft, in conclusion, was a demanding yet satisfying process that called for careful consideration of several variables, including aerodynamics, stability, and manufacturing techniques. We were able to create a practical and effective design that satisfied the competition's criteria and goals through research, analysis, and testing. We were able to achieve adequate performance and control while also guaranteeing maximum payload carrying capacity thanks to the use of efficient materials and technology. Overall, this project shows the value of meticulous planning, close attention to detail, and teamwork in producing excellent design and engineering solutions. Thus, the Drone Design Challenge 2024 played a significant role in total skill development as designers and engineers.